HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and developed **low-latency, high-throughput trading applications** using **modern C++ (C++17)**, focusing on efficient **memory management** and **multi-threading** for real-time data processing.
- Optimized critical code paths in C++ trading systems, reducing **CPU utilization** by an average of **18%** and improving **event processing** speed in a high-frequency environment.
- Implemented inter-process communication (IPC) mechanisms using **Boost.Interprocess** and shared memory segments to facilitate high-speed data exchange between distributed trading components.
- Collaborated with quantitative researchers to translate complex trading algorithms into performant C++ code, contributing to the refinement of strategies and achieving a **15% improvement in execution speed**.
- Engineered **network/socket programming** solutions for low-latency data ingestion and order execution, ensuring reliable data streams in a demanding financial market.
- Participated in full software development lifecycle activities, including **code reviews**, **sprint planning**, and **technical estimation** for new feature development on the trading platform.
- Containerized C++ applications using **Docker** and deployed them to **AWS EC2** instances, establishing basic CI/CD pipelines with **Jenkins** for automated builds and testing.

SKILLS

Languages – C++, Python, Java, SQL

C++ Libraries & Concepts – C++11/14/17/20, STL, Boost (Boost.Interprocess, Boost.ASIO), Multi-threading, Memory Management, IPC, Network/Socket Programming

Performance & Optimization – Low-latency systems, High-throughput systems, Profiling, Performance Tuning, System Bottlenecks

Cloud & DevOps – AWS (EC2, S3, CloudWatch), Docker, Jenkins, Git, Linux, Shell Scripting

Financial Systems – High-frequency trading, Trading Technology, Quantitative Research Collaboration, Real-time Environments

PROJECTS

Low-Latency C++ Market Data Handler

- Developed a C++17 application to process and parse high-volume market data feeds from multiple exchanges, achieving p99 latency below **210ms**.
- Implemented efficient **multi-threading** for concurrent data ingestion and normalization, utilizing **STL** containers and custom allocators for optimized memory access.
- Integrated with an existing C++ trading backend via **Boost.Interprocess** shared memory, ensuring rapid dissemination of critical price updates.

C++ Strategy Optimization Engine

- Built a performance-critical C++ module to backtest and optimize parameters for quantitative trading strategies, reducing simulation times by **30%**.
- Applied profiling tools to identify and resolve bottlenecks in **memory management** and **algorithm execution** within the strategy engine.
- Designed a flexible interface for loading strategy configurations, enabling rapid iteration by quantitative researchers on new trading hypotheses.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)